QAOA no longer generates random initial points

https://github.com/qiskit-community/qiskit-aqua/issues/1417

ROW 60

QAOA no longer generates random initial points #1417

Closed #1424

xenoicwyce opened on Nov 7, 2020

Information

I've been using Qiskit 0.19 for quite some time until I recently upgraded to 0.23. Then I realized my QAOA is not working correctly. I'm not sure whether the problem is related to Aqua or not, but I hope someone can fix this.

The QAOA class in Aqua takes in an `initial_point` parameter. According to the Qiskit docs, a random initial point should be generated when `initial_point=None`, which is `None` by default. However, after I upgraded to Qiskit 0.23, the `QAOA.optimal_params` always returns an array of 0, which is strange. The results is also incorrect.

- **Qiskit Aqua version**: 0.8.0
- **Python version**: 3.8.6
- **Operating system**: Windows 10

What is the current behavior?

When `initial_point=None` is passed into the QAOA class, the `QAOA.optimal_params` always return an array of 0. Apparently, the `initial_point=[0., 0.]` is passed into QAOA instead of a random array.

Steps to reproduce the problem

```
import networkx as nx

from qiskit import BasicAer
from qiskit.optimization.applications.ising import max_cut

from qiskit.aqua import QuantumInstance
from qiskit.aqua.algorithms import QAOA
from qiskit.aqua.components.optimizers import NELDER_MEAD

node = 5
prob = 0.5
seed = 123

p = 1
quantum_seed = 10598
shots = 4096

G = nx.fast_gnp_random_graph(node, prob, seed=seed)
w = nx.adjacency_matrix(G).toarray()

qubit_op, offset = max_cut.get_operator(w)
optimizer = NELDER_MEAD(disp=True)
qaoa = QAOA(qubit_op, optimizer, p) # this is where the problem occurs

backend = BasicAer.get_backend('qasm_simulator')
quantum_instance = QuantumInstance(backend,
                                   seed_simulator=quantum_seed,
                                   seed_transpiler=quantum_seed,
                                   shots=shots)
result = qaoa.run(quantum_instance)

print('Optimal params:', qaoa.optimal_params)
```

Results on Qiskit 0.19:

```
Optimization terminated successfully.
         Current function value: -0.953613
         Iterations: 36
         Function evaluations: 86
Optimal params: [1.14869644 5.63889282]
```

Results on Qiskit 0.23:

```
Optimization terminated successfully.
         Current function value: -0.019043
         Iterations: 3
         Function evaluations: 11
```

Assignees

manoelmarques

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

QAOA should generate random initial points if they are None

qiskit-community/qiskit-aqua

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Participants

Zotero

```
        Iterations: 3
        Function evaluations: 11
Optimal params: [0. 0.]
```

What is the expected behavior?

The `QAOA.optimal_params` should return a non-zero floating point array for most of the time, just as shown above in Qiskit 0.19.

Suggested solutions

My current workout is to downgrade back to 0.19. However, one can also manually pass a random angle into `initial_point` to make the QAOA work correctly again.

xenoicwyce on Nov 7, 2020 Author ···

Sorry, just realized it was a duplicate issue. #1315 #1316

woodsp-ibm on Nov 7, 2020 Member ···

Your issue is certainly related but is a bit different. QAOA var form always indicated a preferred_init_points of 0's, which is still the case.
One thing which did change after 0.19 was released is that we changed the random number generation in Qiskit to use the newer numpy function for the random number generation. This no longer generates the same random numbers for the same seed, but I do not know if this is impacting your sample or not. Its something that needs investigating for sure so thanks for bringing this to our attention and providing your code sample.

xenoicwyce on Nov 9, 2020 Author ···

Pretty sure I only seeded for the simulator and the transpiler. Does this affect the random initial points?

manoelmarques self-assigned this on Nov 9, 2020

woodsp-ibm on Nov 9, 2020 Member ···

VQE (and QAOA) take an `initial_point` which defaults to None if not supplied. So if you supply one it uses it. Otherwise with the default it first checks the variational form to see if it has a preferred initial point and if not will compute a random one. Now the QAOA var form has a preferred initial point of all zeros - it always has had this. Hence QAOA should not be computing a random initial point. But of course throughout qiskit, the random number is now generated a different way, which will affect the transpiler and simulator and hence the same seed will now have a different behavior. I would have hoped the random sampling from the distribution would not have affected things where perhaps it gets stuck. It always worked with NELDER_MEAD I guess - did you try COBYLA. Anyway we will look into what is happening.

xenoicwyce on Nov 10, 2020 Author ···

I've tried with a few configurations. Not seeding the quantum instance seems to solve the problem for COBYLA, but will cause NELDER_MEAD to exceed the max function evaluation (I think that's the reason I seeded the quantum instance at the first place). However, in 0.19, even if the quantum instance is seeded, it still gives different values every run, which is not the case in 0.23.

The results are shown below.

Qiskit 0.19.0 (Aqua 0.7.0):

```
Trying with COBYLA, seeding quantum instance...
```

```
Trying with COBYLA, seeding quantum instance...

Optimal params: [1.18135966 5.68222941] # this value is different for every run
---

Trying with COBYLA, without seeding quantum instance...

Optimal params: [1.17610763 2.48016696]
---

Trying with NELDER_MEAD, seeding quantum instance...
Optimization terminated successfully.
         Current function value: -0.718506
         Iterations: 45
         Function evaluations: 96

Optimal params: [ 5.07512673 -0.72556979] # different every run
---

Trying with NELDER_MEAD, without seeding quantum instance...
Warning: Maximum number of function evaluations has been exceeded.

Optimal params: [1.09503068 5.64349897]
```

Qiskit 0.23.0 (Aqua 0.8.0):

```
Trying with COBYLA, seeding quantum instance...

Optimal params: [0. 0.]
---

Trying with COBYLA, without seeding quantum instance...

Optimal params: [0.68788458 1.29580419]
---

Trying with NELDER_MEAD, seeding quantum instance...
Optimization terminated successfully.
         Current function value: -0.005371
         Iterations: 3
         Function evaluations: 11

Optimal params: [0. 0.]
---

Trying with NELDER_MEAD, without seeding quantum instance...
Warning: Maximum number of function evaluations has been exceeded.

Optimal params: [8.30078125e-05 1.34765625e-04]
```

xenoicwyce on Nov 10, 2020 Author ···

Code

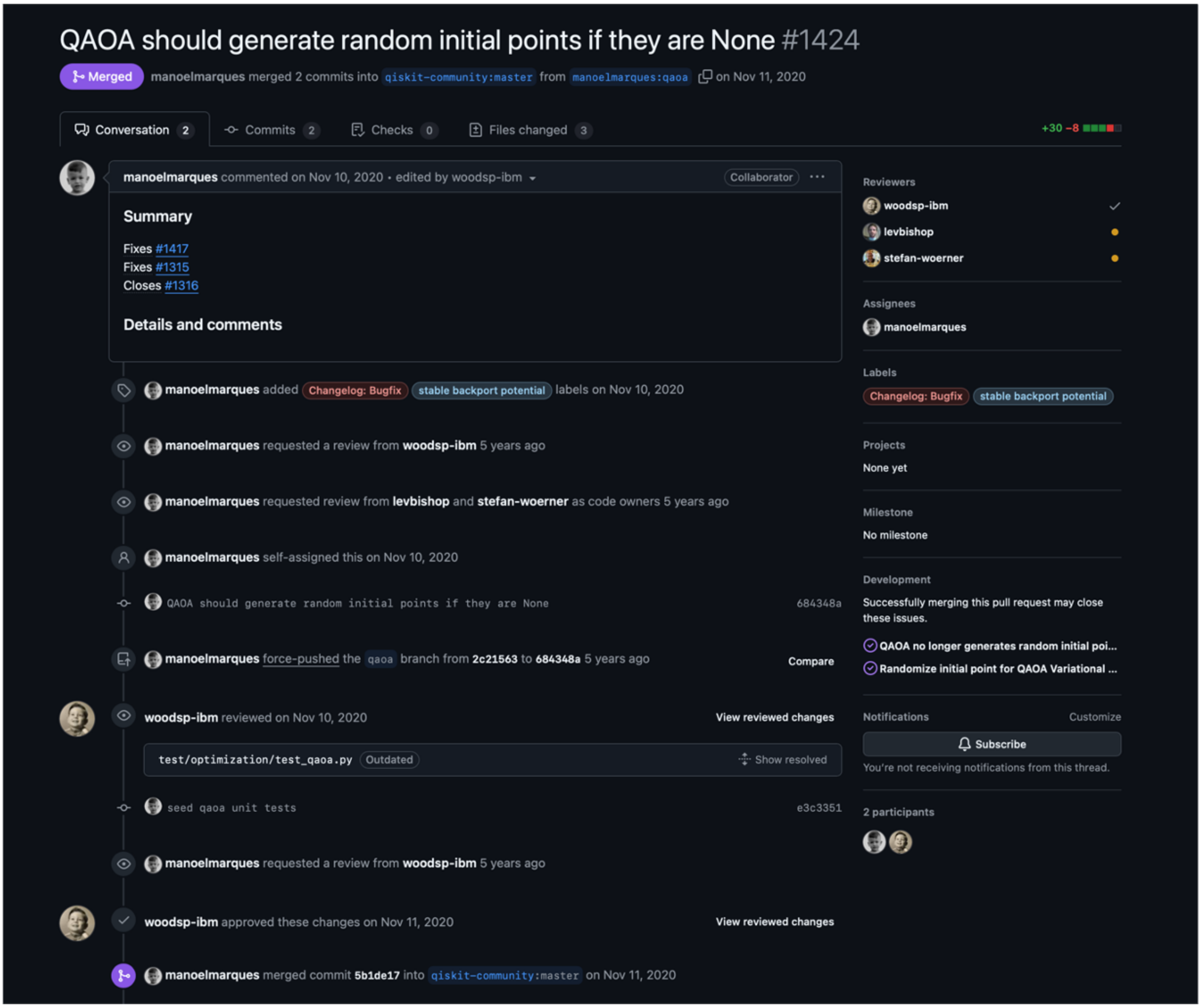
```
import qiskit

print('Qiskit version:', qiskit.__qiskit_version__['qiskit'])
print('Aqua version:', qiskit.__qiskit_version__['qiskit-aqua'])
print('')

import networkx as nx

from qiskit import BasicAer
from qiskit.optimization.applications.ising import max_cut

from qiskit.aqua import QuantumInstance
from qiskit.aqua.algorithms import QAOA
from qiskit.aqua.components.optimizers import NELDER_MEAD, COBYLA

import warnings
warnings.filterwarnings('ignore')

node = 5
prob = 0.5
```

```
import qiskit

print('Qiskit version:', qiskit.__qiskit_version__['qiskit'])
print('Aqua version:', qiskit.__qiskit_version__['qiskit-aqua'])
print('')

import networkx as nx

from qiskit import BasicAer
from qiskit.optimization.applications.ising import max_cut

from qiskit.aqua import QuantumInstance
from qiskit.aqua.algorithms import QAOA
from qiskit.aqua.components.optimizers import NELDER_MEAD, COBYLA

import warnings
warnings.filterwarnings('ignore')

node = 5
prob = 0.5
seed = 123

p = 1
quantum_seed = 10598
shots = 4096

seed_quantum_instance = True

G = nx.fast_gnp_random_graph(node, prob, seed=seed)
w = nx.adjacency_matrix(G).toarray()

qubit_op, offset = max_cut.get_operator(w)
optimizer = NELDER_MEAD(disp=True)
msg = '' if seed_quantum_instance else 'without '
print(f'Trying with {type(optimizer).__name__}, {msg}seeding quantum instance...')

qaoa = QAOA(qubit_op, optimizer, p)

backend = BasicAer.get_backend('qasm_simulator')
if seed_quantum_instance:
    quantum_instance = QuantumInstance(backend,
                                       seed_simulator=quantum_seed,
                                       seed_transpiler=quantum_seed,
                                       shots=shots)
else:
    quantum_instance = QuantumInstance(backend,
                                       shots=shots)
result = qaoa.run(quantum_instance)

print('\nOptimal params:', qaoa.optimal_params)
```

manoelmarques mentioned this on Nov 10, 2020

QAOA should generate random initial points if they are None #1424

manoelmarques closed this as completed in #1424 on Nov 11, 2020



QAOA should generate random initial points if they are None #1424

Merged manoelmarques merged 2 commits into `qiskit-community:master` from `manoelmarques:qaoa` on Nov 11, 2020

Conversation 2 | Commits 2 | Checks 0 | Files changed 3 | +30 −8

manoelmarques commented on Nov 10, 2020 • edited by woodsp-ibm | Collaborator

Summary

Fixes #1417
Fixes #1315
Closes #1316

Details and comments

manoelmarques added Changelog: Bugfix stable backport potential labels on Nov 10, 2020

manoelmarques requested a review from **woodsp-ibm** 5 years ago

manoelmarques requested review from **levbishop** and **stefan-woerner** as code owners 5 years ago

manoelmarques self-assigned this on Nov 10, 2020

`QAOA should generate random initial points if they are None` 684348a

manoelmarques force-pushed the `qaoa` branch from **2c21563** to **684348a** 5 years ago — Compare

woodsp-ibm reviewed on Nov 10, 2020 — View reviewed changes

`test/optimization/test_qaoa.py` Outdated — Show resolved

`seed qaoa unit tests` e3c3351

manoelmarques requested a review from **woodsp-ibm** 5 years ago

woodsp-ibm approved these changes on Nov 11, 2020 — View reviewed changes

manoelmarques merged commit **5b1de17** into `qiskit-community:master` on Nov 11, 2020

Reviewers
- woodsp-ibm
- levbishop
- stefan-woerner

Assignees
- manoelmarques

Labels
- Changelog: Bugfix
- stable backport potential

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
- QAOA no longer generates random initial poi...
- Randomize initial point for QAOA Variational ...

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2 participants